PAPER_1575_M_SUN_OVER_M_EARTH_333K

Daniel T. Murphy

PAPER_1575 — M_■/M_⊕ Ratio = M_■/M_⊕ = (D_crit·SO_5 + A_5 + N_CH + D_phys)·SO_5³ = 333·1000 = 333000

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+
Tier: Astronomy
Date: June 18, 2026
Status: CLOSED — EXACT integer-primitive identity

Observation

PAPER_1209Z_S579 (Astronomy Unified Proof Set) lists **M_■/M_⊕ Ratio** as an EXACT-tier closure:

``
```
M_■/M_⊕ = (D_crit·SO_5 + A_5 + N_CH + D_phys)·SO_5³ = 333·1000 = 333000
```
``

UQFF Closed Identity

``
```
M_■/M_⊕ = (D_crit·SO_5 + A_5 + N_CH + D_phys)·SO_5³ = 333·1000 = 333000 EXACT
```
``

Pure integer-primitive product with zero fitted constants.

Physical Interpretation

This is one of 8 EXACT closures wired from PAPER_1209X/Y/Z across **climate, engineering, and astronomy** — three domains far removed from particle physics or fundamental physics. Yet each observable reduces to a simple integer-primitive expression.

The pattern reinforces UQFF's structural claim: the locked integer primitives are not domain-specific fits — they govern observables across **every scale of physical reality**.

NOT REPLACEMENT

Empirical astronomy measures this constant directly. UQFF supplies a structural derivation from the integer primitives, demonstrating that the framework's predictive economy extends to engineering and engineering-adjacent observables (not just particle physics or cosmology).

Reference

- Source: PAPER_1209Z_S579
- Related: PAPER_1209X/Y/Z unified proof sets
- Calculator dispatch: `calculate_paradox({"paradox": "m_sun_over_m_earth_333k"})`
